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CLARIFICATIONS OF THE BCU METHOD FOR TRANSIENT STABILITY ANALYSIS

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ABSTRACT

Energy function methods have been studied for many years, and have been applied to practical power system stability analysis problems of multi-machine power systems. Recent developments in real-time power system monitoring suggest that dynamic events can be monitored at the power system control centers, and naturally the energy function methods were tried as real-time stability prediction tools. However, a number of instances were uncovered, where the energy function methods which use the Potential Energy Boundary Surface as an approximation of the stability boundary produce unreliable results. In particular, during several transient stability studies, the Boundary Controlling Unstable (BCU) Equilibrium Point method seemed to predict stable swings, whereas in reality the swings turned out to be unstable. This paper presents these counter-examples, and suggests an explanation as to why these methods produce a wrong result. It is hoped that this paper will lead to further researches and improvements in the theory of energy function based methods of stability analysis. In the mean time, alternative methods for the real-time stability prediction problems are under investigation.

1. INTRODUCTION

Our interest in the direct methods of transient stability analysis developed from the problem of real-time prediction of instability and control. It has become possible to contemplate such tasks because of the newly developed technique of synchronized phasor measurements [1] and their use in adaptive out-of-step relaying and other real-time control functions.

We began by considering the Energy Function Methods, and in particular their various versions which use the Potential Energy Boundary Surface (PEBS) as an approximation of the actual system stability boundary. These methods have been described extensively in the literature [2,3,4,5,6]. As frequently stated, these methods are expected to predict only the first swing stability of the multi-machine power system. It is often felt that this may be a good enough statement about the stability. In reality, the result only asserts that there will be a first turning point in the oscillation of the rotors. In fact this statement is unsatisfactory, as the oscillation could (and does) go unstable in the second half of the electromechanical swing. This effect is sometimes described as instability on the second swing. For all practical purposes, a system that goes unstable on the second swing is unstable, and to know that it had a first turning point is of no help.

More recently, the BCU method has been proposed [6,7], and as part of this development, an important theorem has been stated, which relates the stability boundary of the actual dynamic system, and that of a half-order (static) load flow system. In particular, it is proved that if certain conditions are satisfied, the unstable equilibria on the stability boundary of the load flow system, and on that of the dynamic system are identical. Consequently, one is led to believe that when the conditions of the theorem are satisfied, the stability boundary of the dynamic system could be determined by determining the stability boundary of the load flow system. If this is correct, determining that a certain swing is stable by the BCU method (or by any other method which identifies the controlling UEP) would guarantee that the swing is ***absolutely*** stable. One would then be free of the first swing limitation inherent in much of the earlier work.

In reality, it is found that this is not the case for a surprisingly large number of cases. The BCU method declares several cases to be stable, while in reality they are not. Some become unstable on the second swing, while others go unstable after a few oscillations. In any case, there is the question as to why the methods based upon the BCU technique seem to produce results which are at variance with time domain simulations.

In the following section, we will provide two counter-examples, which show that the predictions of the BCU type method are at variance with the time domain simulations. In the next section, we discuss the flaw in the application of the theory, which has led to this failure. We conclude with some thoughts on what could be done to determine the stability of the

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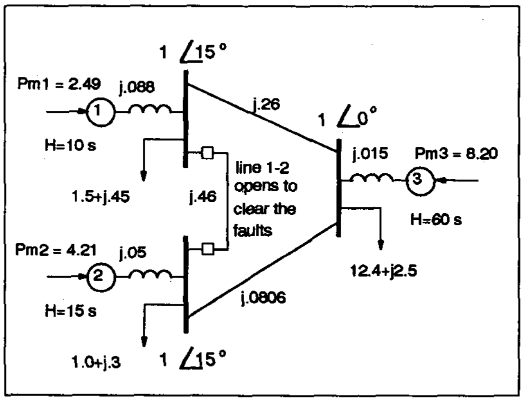
multi-machine power system in general, and in particular in real-time applications.

2. COUNTER-EXAMPLES

Counter-example I



Three-machine system of reference [3].

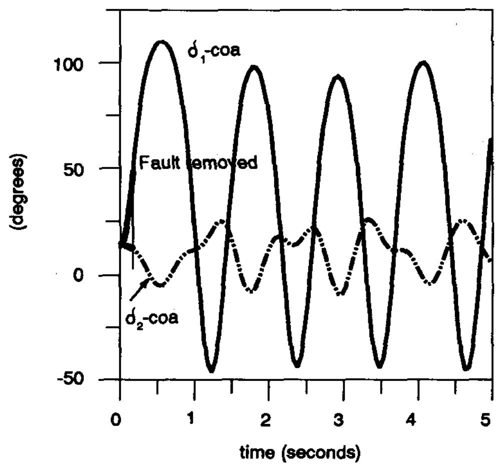
Figure 1

A uniform damping of 0.02 is assumed for the three-machine system [3] shown in Figure 1. Four faults on line 1-2 are considered, and in all cases the fault is isolated by the opening of this line. The boundary controlling unstable equilibria are found by the BCU method, and are listed in the Center of Angle (COA) reference frame in Table 1. Two angles given for the UEP are δ_1 and δ_2 in the COA reference frame. Small values of fault impedances are used (rather than zero impedance faults) for ease of calculation. Fault locations, fault impedances, and the critical clearing time (CCT) obtained by the BCU method are as follows:

Table 1

Fault	At bus	Fault Imp. (pu)	UEP	CCT(s)
(i)	1	j5E-6	(123°,-0.9°)	0.1740
(ii)	1	0.06	(-196.3°,41.5°)	0.3250
(iii)	2	j5E-6	(-13.5°,119.6°)	0.1995
(iv)	2	0.03	(50.0°,-176.9°)	0.4550

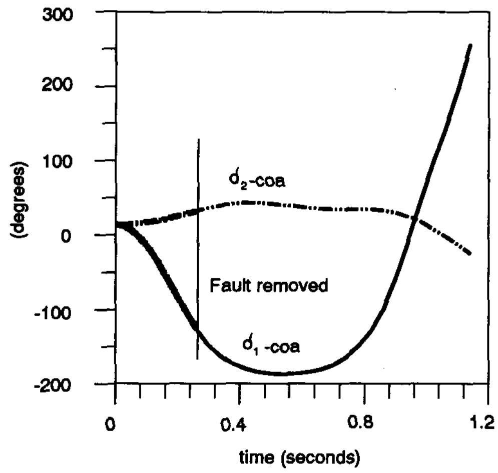
The results of the angles in the COA reference frame vs time, corresponding to the aforementioned faults are shown in Figures 2 through 5. From Figure 2 we can see that the power system is stable when fault (i) is cleared in 0.1740 s. Fault (ii) has the same location as fault (i), but it causes machine one to decelerate. The critical clearing time obtained by the BCU method for this case is 0.3255 s. Figure 3 shows this clearing time to be in error since machine one goes unstable on the



Angles in COA versus time corresponding to fault (i).

Figure 2

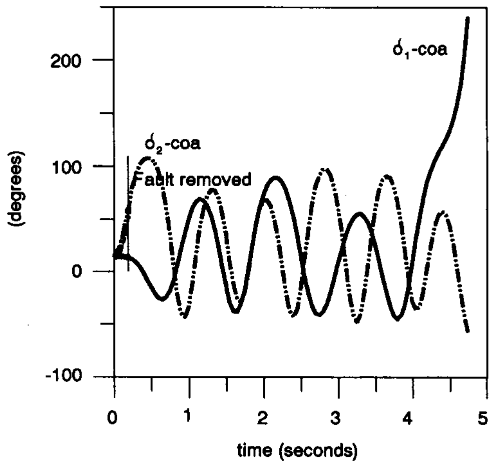
second swing. Figure 4 corresponds to fault (iii), in which case the swing exhibits one mode of oscillation in the beginning (machine two against the other two) and a second mode at a later time (machine one against the other two). Although initially machine two seems to be going unstable, the one that actually goes unstable is machine one. Figure 5 corresponds to fault (iv), which causes machine two to decelerate. However, with the clearing time obtained by the BCU method it goes unstable on the second swing.



Angles in COA versus time corresponding to fault (ii).

Figure 3

Thus, of the four faults studied in this example, three produce instability, whereas the BCU method predicts stability for all the cases.



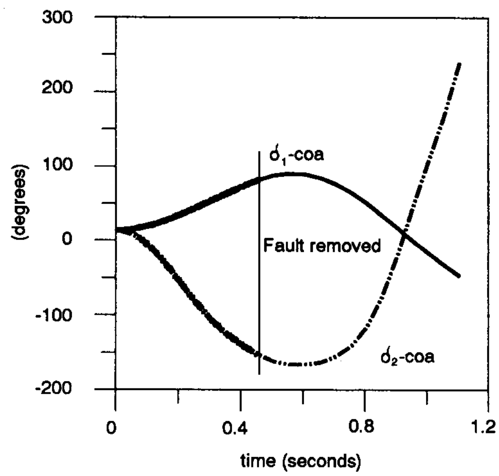
Angles in COA versus time corresponding to fault (iii).

Figure 4

Counter-example II

Let us consider the one-machine infinite-bus system shown in Figure 6. The system parameters are as follows:

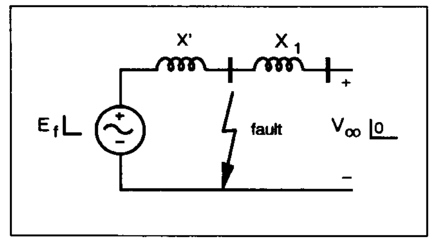
$E_f = 1.05$ pu
$V_\infty = 1.00$ pu
$X' = 0.10$ pu
$X_1 = 0.25$ pu



Angles in COA versus time corresponding to fault (iv).

Figure 5

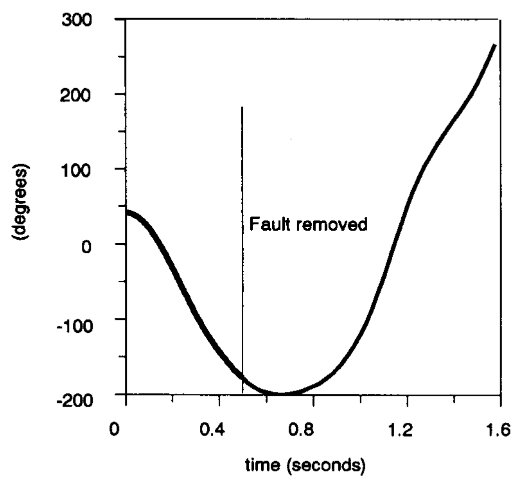
$M = 0.05$ MW s^2/(MVA rad)
$P_m = 2.00$ pu
$D = 0.15$ MW s/(MVA rad)
$Z_f = 0.075$ pu
$P_{max} = E_f V_\infty / (X' + X_1) = 3.0$ pu



One Machine Infinite Bus (OMIB) system.

Figure 6

In this case the uniform damping is $\lambda_d = D/M = 3.0$. The results of the BCU method provide the UEP as -3.8713 rad or -221.8°. The critical clearing time for the fault is 0.50 s. Figure 7 is a plot of the angle versus time corresponding to this critical clearing time, and shows that it is in error since the machine goes unstable on the second swing.



Angle versus time for a clearing time of 0.50 s

Figure 7

3. CLARIFACTIONS

The reason for the failure can be explained with the help of the OMIB system of the previous section. The BCU method is a controlling UEP method, which finds the controlling UEP of the original system by detecting the PEBS crossing and finding a UEP of the half-order gradient system. For the OMIB case, the dynamical equations of the original system and the gradient system are as follows:

$$d_p(M,D): \quad \begin{cases} \dot{\delta} = \omega \\ M\dot{\omega} = P_m - P_{max} \sin(\delta) - D\,\omega \end{cases}$$

$$d_p(1): \quad \left\{ \dot{\delta} = P_m - P_{max} \sin(\delta) \right.$$

The stable equilibrium points (SEPs) of the original system and the gradient system are as follows:

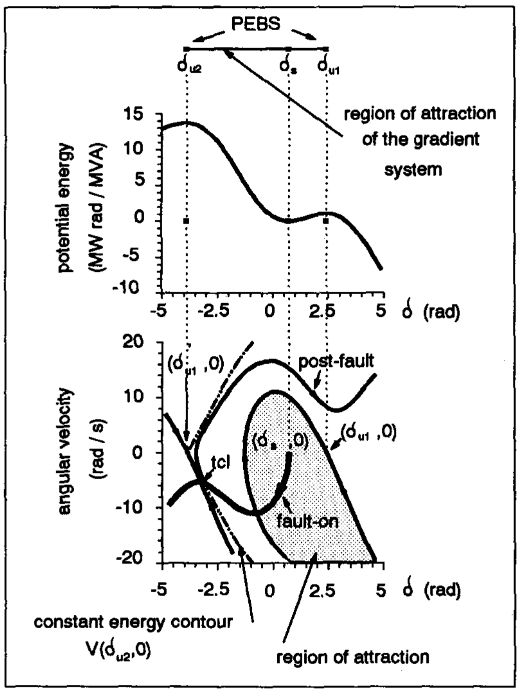
$$\text{Original System:} \quad \begin{cases} \delta_s = \arcsin\left[\dfrac{P_m}{P_{max}}\right] \\ \omega_s = 0 \end{cases}$$

$$\text{Gradient System:} \quad \left\{ \delta = \arcsin\left[\dfrac{P_m}{P_{max}}\right] \right.$$

There are two UEPs surrounding the SEP as follows:

$$\text{Original system:} \quad \begin{cases} \delta_{u1} = \pi - \delta_s \\ \omega_{u1} = 0 \end{cases} ; \text{ and } \begin{matrix} \delta'_{u1} = -\pi - \delta_s \\ \omega'_{u1} = 0 \end{matrix}$$

$$\text{Gradient system:} \quad \left\{ \delta_{u1} = \pi - \delta_s; \text{ and } \delta_{u2} = -\pi - \delta_s \right.$$



Regions of stability of the full and half order systems for OMIB.
Figure 8

The equilibrium points of both the original system and the gradient system are shown in Figure 8, and the region of attraction of $(\delta_s,0)$ is shown by the shaded area. The region of attraction of (δ_s) is the open interval $\delta_{u2} < \delta < \delta_{u1}$. The stability boundary of $(\delta_s,0)$ is the stable manifold of $(\delta_{u1},0)$, which is the line surrounding the shaded area. The stability boundary of the gradient system (PEBS) in this case is $\delta_{u2} \cup \delta_{u1}$.

Using the BCU method, the maximum of potential energy along the fault-on trajectory is detected (the fault-on trajectory projected onto the δ-axis intersects with PEBS at the PEBS exit point t_{cl}) to find the UEP, δ_{u2} in this case. Then, under the assumption of ***the one-parameter transversality condition*** [6], the constant energy surface given by $V(\delta_{u2},0)$ is used as the local approximation of the stability boundary of the SEP $(\delta_s,0)$. We can see from Figure 8 that in this case the local approximation is in error. The UEP (δ_{u2}) is on the PEBS, but the UEP $(\delta'_{u1},0)$ is not on the stability boundary of $(\delta_s,0)$. As we will show next, the assumption of the one-parameter transversality condition is not valid in this case.

One-parameter transversality condition:

As in [6, 7], we consider the following two dynamical systems:

(1) ***The changing $-D_\lambda^{-1}$ gradient system,***
This is a half-dimensional system,

$$d_p(D_\lambda^{-1}): \quad \left\{ \dot{\delta} = D_\lambda^{-1} \left[P_m - P_{max} \sin(\delta)\right] \right.$$

where $D_\lambda^{-1} = \lambda(D^{-1} - 1) + 1$, and $\lambda = [0, 1]$.
This system differs from the gradient system only because of D_λ^{-1} which changes from 1 to D^{-1} as λ changes from 0 to 1.

(2) ***The changing-inertia system,***

$$d_p(M_\lambda,D): \quad \begin{cases} \dot{\delta} = \omega \\ M_\lambda \dot{\omega} = P_m - P_{max} \sin(\delta) - D\omega \end{cases}$$

where $M_\lambda = \lambda M + (1 - \lambda)\varepsilon$, where ε is suitably small, and $\lambda = [0, 1]$. This system is the same as the original system for $\lambda = 1$.

The ***one-parameter transversality condition*** is considered to be satisfied if the two dynamical systems defined above, $d_p(M_\lambda,D)$ and $d_p(D_\lambda^{-1})$ satisfy the following assumptions for $0 \le \lambda \le 1$:

C1 **(*Transverse intersection*)** Either the stable and the unstable manifolds of the equilibria on the stability boundary do not intersect each other, or the intersections satisfy the transversality condition.

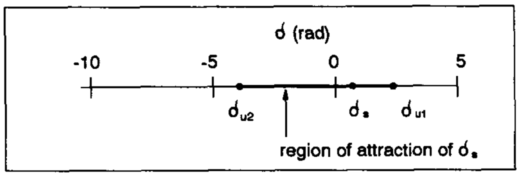
C2 (*Finite number of points on the stability boundary*) The number of equilibria on the stability boundary be finite.

OMIB system and the one-parameter transversality:

Using the parameters given in counter-example II for the OMIB, we will now check if the one parameter transversality condition is satisfied by this system:

(1) ***The changing D_λ^{-1} gradient system:***

$$d_p(D_\lambda^{-1}): \quad \left\{ \dot{\delta} = D_\lambda^{-1}[2 - 3\sin(\delta)] \right.$$



The system $d_p(D_\lambda^{-1})$ satisfies the one parameter transversality condition.

Figure 9

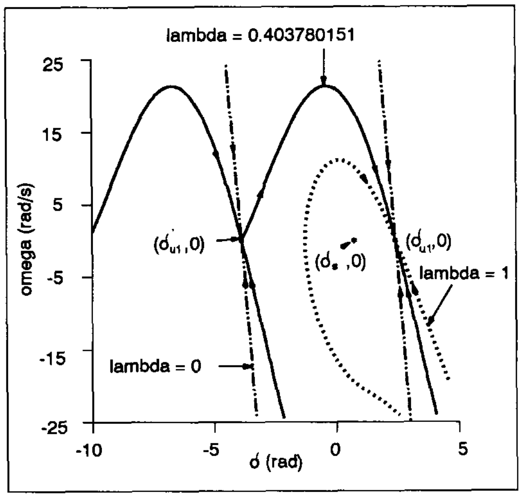
This system has the same equilibria as the gradient system. The region of attraction of (δ_s) is the open interval (δ_{u2}, δ_{u1}) as shown in Figure 9. Neither the region of attraction, nor the equilibria change with λ. There are two equilibria on the stability boundary (δ_{u2}, δ_{u1} are the stability boundary) and they do not intersect with each other. Therefore, this system satisfies conditions C1 and C2.

(2) ***The changing-inertia system,***

$$d_p(M_\lambda, D): \quad \begin{cases} \dot{\delta} = \omega \\ M_\lambda \dot{\omega} = 2\text{-}3\sin(\delta)\text{-}0.15\omega \end{cases}$$

$$M_\lambda = 0.05\,(0.9\,\lambda + 0.1) = \begin{cases} 0.005 \text{ if } \lambda = 0 \\ 0.05 \quad \text{if } \lambda = 1 \end{cases}$$

where, for convenience, we have selected $\varepsilon = 0.1M$. This system has the same equilibria as the original system. The equilibria do not change with λ, but the region of attraction does. Figure 10 shows the stability boundary, but for $\lambda = 0.403780151$ the system does not satisfy the transversality condition. There is a saddle connection, i.e. part of the unstable manifold of (δ'_{u1}, 0) is also a part of the stable manifold of (δ_{u1}, 0), and at this intersection the tangent space is one-dimensional. ***Therefore, this system satisfies C2, but not C1.*** Consequently, since $d_p(M_\lambda, D)$ does not satisfy condition C1, the original OMIB system does not satisfy the one-parameter transversality condition.



The dynamical system dp(M$_\lambda$,D) does not satisfy the one-parameter transversality condition.

Figure 10

As mentioned before, the BCU method is a technique for finding the controlling UEP. It should also be recognized that in a lossy system the potential energy is path dependent, and several techniques have been proposed to approximate the path dependent energy function, such as the linear trajectory approximation [3], and the trapezoidal rule approximation [8]. These approximations in turn may affect the ability of the BCU method to find the controlling UEP in practical lossy systems.

A check for the validity of the BCU method:

When the one-parameter transversality condition is satisfied, the number of equilibrium points on the PEBS is the same as the number of equilibrium points on the stability boundary of the original system. When it is not satisfied, the number of equilibrium points on the PEBS seems to be greater than the number of equilibrium points on the stability boundary of the original system.

Failure of the assumption of the one-parameter transversality condition may lead the PEBS and BCU methods into giving incorrect results for multi-swing stability. To verify the one-parameter transversality condition by varying λ for a multi-machine system as we did for the OMIB system is an impossible task. However, we can verify directly if a UEP determined by the BCU method ($\underline{\delta}_{uep}$, $\underline{0}$) lies on the stability boundary of the original system, in the following manner [9]:

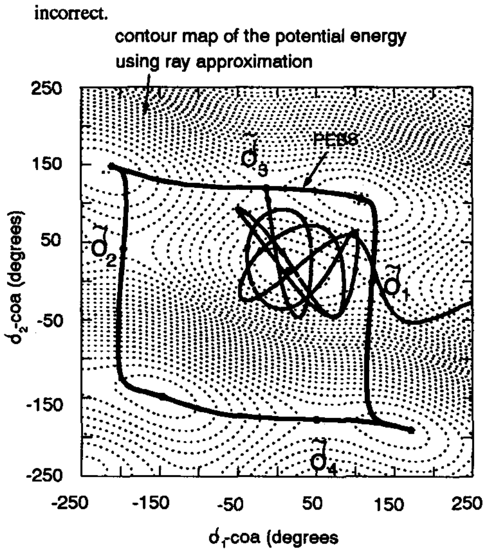
(1) Define a normalized vector $\underline{z}$ as follows:

$$\underline{z} = \frac{\underline{\delta}_s - \underline{\delta}_{uep}}{\|\underline{\delta}_s - \underline{\delta}_{uep}\|}$$

(2) Integrate the post-fault dynamic equation of the original system ***along a test trajectory*** using a starting point just inside the estimated stability boundary: $(\underline{\delta}_{uep} + 0.01\underline{z},\underline{0})$. There are two possible outcomes of this calculation:

(a) If this test trajectory tends towards $(\underline{\delta}_s,\underline{0})$ then the point $(\underline{\delta}_{uep},\underline{0})$ lies on the stability boundary of the original system. In this case, the stability boundary may be locally approximated by the constant energy trajectory $V(\underline{\delta}_{uep},\underline{0})$ and the critical clearing time may be approximated by the time at which the fault-on trajectory intersects with this surface. The prediction of the BCU method is correct for this case.

(b) If the test trajectory tends towards infinity, then the point $(\underline{\delta}_{uep},\underline{0})$ does not lie on the stability boundary of the original system, and the one-parameter transversality condition is not met. The prediction of the BCU method is incorrect.



Trajectory projected onto the angle subspace.

Figure 11

Such a test has been applied to case (iii) of counter-example I. The result is shown in Figure 11. It can be seen that as $t \to \infty$, the trajectory escapes the neighborhood of the stable equilibrium point, and the case is unstable. Note that this test is not of much practical use, in that the test of the validity of the BCU method itself involves the same amount of computation as needed to solve the original system of the equations in time domain.

4. CONCLUSIONS

(1) We were motivated by the potential for real time application of stability prediction methods, which could use real time synchronized phasors obtained by direct measurement from the power system. To this end, in the interest of finding a computationally efficient method, we explored various versions of the energy function based methods of stability assessment, the most recent version of these methods being the BCU method.

(2) We find that although the predictions of the BCU method are correct in many cases, in many other cases the BCU method declares the system stable, while in reality it is unstable. All of these cases of failures occur after the rotor angles have turned around once. Thus, one could loosely say that the predictions of the first swing stability are correct, if one were to interpret the first swing to mean the first turning point of the oscillation. However, in terms of stability assessment, this limited view of stability is of no practical use.

(3) A closer investigation of the cases when the predictions of the BCU method fail, reveals that the failure is caused by the system not satisfying the one-parameter transversality condition defined in the literature on the mathematical basis of the BCU method [7]. We find that to ensure that the prediction of the BCU method is correct, one must check the stability of the trajectory which begins inside the stability boundary, arbitrarily close to the controlling unstable equilibrium point. Although this test seems to be conclusive, it is of little practical help.

(4) We anticipate further theoretical developments on the part of the research community which would provide a reliable estimate of the stability based upon the energy function methods. In the mean time, we are investigating alternative procedures to assess stability of power systems in real time applications.

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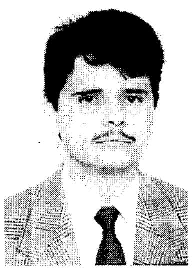

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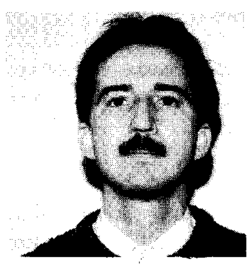

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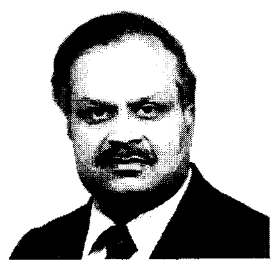

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Discussion

G. C. Ejebe and J. Z. Tong (Empros Power Systems Control, Plymouth, MN): The authors are to be commended for their interesting paper which presents their difficulties with the BCU method for transient stability analysis. We read this paper with a lot of interest since we and our colleagues at Empros in collaboration with Iowa State University have been developing a Sparse Transient Energy Function (STEF) program for contingency screening and ranking for on-line dynamic security analysis as part of an EPRI-sponsored project. STEF is based on the boundary of stability region based controlling UEP method (BCU).

We are puzzled by the results shown in Table 1 of the paper and on which the conclusions about the apparent failure of the BCU method in three cases have been drawn.

We therefore decided to run the tests on the 3-machine system using our STEF program. This program computes the controlling UEP [1] as the UEP whose stable manifold contains the exit point of the stability boundary of the reduced gradient system. The overall solution procedure consists of the following computational steps:

- Computer of specified fault trajectory to fault clearing. (Any fault impedance can be specified.)
- Computation of the exit point using sustained *short circuit* fault-on trajectory
- Computation of the Minimum Gradient Point (MGP)
- Computation of the controlling UEP
- Computation of the energy margin

Table A shows the results obtained using STEF as well as performing time domain simulation using EPRI's ETMSP Version 3.0 program. The discrepancy between these results and those reported in the paper is presumably due to some small but *significant* algorithmic details in the respective programs. In the STEF program, the exit point is determined by performing short circuit (j1.0E-12) fault-on trajectory computation until the potential energy of the post-disturbance network reaches its first local maximum. Thus, given a fault at a particular bus, the same controlling UEP is obtained, regardless of the fault impedance value and duration, provided the post-fault network is the same!

As can be seen from Table A, STEF gives very good assessment for these cases. In fact, STEF has been successfully applied to practical-sized power systems with up to 250-generators.

Table B shows the driving point (Thevenin) impedance for the two buses of interest, computed without load and with loads modeled as constant impedances. In view of these values, why did the authors choose such high values (0.06 or j0.06) for fault impedances, especially at Bus 2? It does appear that a value of "fault impedance" of j0.06 at Bus 2 is not realistic. As seen from Table A, the system is absolutely stable if this "fault impedance" is applied at Bus 2, no matter how long it is left on before clearing!

Thus, it appears to us that the author's difficulties with the counter examples stem from a combination of two issues:

a. Their selection of large unrealistic fault impedances
b. Their attempt to trace a sustained fault trajectory to the exit point with unrealistic fault impedance inevitably leads to the wrong exit point and UEP.

TABLE A
RESULTS FROM STEF AND ETMSP.

Fault	Bus Location	Fault Impedance (PU)	UEP	CCT(s) (ETMSP)	CCT(s) (STEF)
(i)	1	j5E-6	(121.9°, -0.9°, - 20.1°)	0.177	0.175
(ii)	1	j.0.06	(121.9°, -0.9°, -20.1°)	0.395	0.385
(iii)	2	j.5E-6	(-18.4°, 120.5°, -27.1°)	0.192	0.195
(iv)	2	0.06	(-18.4°, 120.5°, -27.1°)	0.192	0.179
(v)	1	0.06	(121.9°, -0.9°, -20.1°)	0.125	0.120
(vi)	2	j.0.06	(-18.4°, 120.5°, -27.1°)	sss	sss
(vii)	2	j.0.03	(-18.4°, 120.5°, -27.1°)	0.650	0.650

sss = absolutely stable

TABLE B
DRIVING POINT IMPEDANCES.

Bus Location	Without Load	With Load Impedance
Bus 1	j0.059	0.0033 + j0.0463
Bus 2	j0.031	0.0012 + j0.0305

Reference

[1] G. D. Irisarri, G. C. Ejebe, J. C. Waight, and W. F. Tinney, "Efficient Solution for Equilibrium Points in Transient Energy Function Analysis" Proceedings of 1993 PICA Conference; Scottsdale, Arizona, pp 426–632, May 1993.

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Claudio A. Cañizares (University of Waterloo): This interesting paper addresses one of the shortcomings of Energy Function methods for directly assessing stability of power systems. Energy Function techniques have several limitations, most of them related to the difficulty of defining global Lyapunov Functions for complex system models. In realistic cases the most one can hope for is to obtain local Energy Functions that allow to produce relative stability estimates of power systems (e.g., [A1]). This problem has force researchers to make several simplifications and general assumptions regarding system modeling, such as assuming lossless systems or using rather simple load models [A2]. When these hypotheses fail, the methods also fail, as demonstrated by the analysis in the paper of the BCU technique, and also by the discussion in [A3] from the original designers of the BCU method when explaining critical clearing time overestimates for a test case. One must also remember that most of these methods are based in Lyapunov Stability theory, which only gives sufficient conditions of stability, i.e., when the energy test fails one cannot draw definite conclusions regarding the stability of the system [A4]. The paper discusses the particular problem of second swing instabilities, and the authors study the problem in detail for the one-generator-infinite-bus case (OMIB in the paper), based on the transversality conditions required by the BCU method. Nevertheless, this discusser would like to analyze the problem from a different perspective.

The second swing instability seems to appear in the presented examples due to the small value of damping used. Notice that in realistic system models, the damping is significantly large due to the direct or indirect modeling of damper windings; therefore, the problem seldom arises in practical applications. This can be easily justified in the OMIB case. Thus, consider zero damping D and a fault-on trajectory moving in the direction of the second u.e.p. $\delta_{u_2} = -\pi - \delta_s$, similar to the case shown in Fig. 7 in the paper. When the system reaches a value of $\delta < \delta_c = -\delta_{u_1} + 2\delta_s$, the trajectory crosses the boundary $\partial A(\delta_s)$ of the stability region $A(\delta_s)$ of the corresponding s.e.p. δ_s ($\delta_c \in \partial A(\delta_s)$). In this case, $\partial A(\delta_s)$ corresponds to the stable manifold $W^s(\delta_{u_1})$ of the first u.e.p. $\delta_{u_1} = \pi - \delta_s$, which is similar to the example depicted in Fig. 8 in the paper. Hence, when the fault is cleared, not enough energy can be dissipated on the swing back to recover stability, and the trajectory leaves the region of attraction "through" the first u.e.p. δ_{u_1} in a second swing. This happens well before the fault-on trajectory crosses the stable manifold $W^s(\delta_{u_2})$ of δ_{u_2}, and reaches a maximum of the potential energy function

$$V_P(\delta) = P(\delta_s - \delta) + \frac{E_f V_\infty}{X' + X_1}[\cos(\delta_s) - \cos(\delta)]$$

According to this stability analysis the controlling u.e.p. should be δ_{u_1} for this particular example, but the BCU method as stated is not able to detect it, since its first step is based on finding a local maximum of V_P. As the value of D increases, then $|\delta_c|$ increases too, until it reaches the value $\delta_c = \delta_{u_2}$ for a "critical" damping D_c. This corresponds to the limit case shown in Fig. 10 of the paper, where part the unstable manifold of δ_{u_2} is also part of the stable manifold of δ_{u_1}, i.e., $W^u_-(\delta_{u_2}) = W^s_+(\delta_{u_1})$. Hence, for values of $D > D_c$, the transversality conditions of the BCU method apply, and the technique is then able to detect the "correct" controlling u.e.p. This brief analysis corroborates the observations of the authors on the second swing problem. However, notice that the stability theory in which the BCU method is based, and that defines the boundaries of $A(\delta_s)$, holds for any value of D; the problem lies on the proposed practical methodology to find $\partial A(\delta_s)$.

The authors' views on these issues would be appreciated.

[A1] C. L. DeMarco and C. A. Cañizares, "A Vector Energy Function Approach for Security Analysis of AC/DC Systems," *IEEE Transactions on Power Systems*, Vol. 7, No. 3, August 1992, pp. 1001-1011.

[A2] M. A. Pai, *Energy Function Analysis for Power System Stability*, Kluwer Academic Publishers, 1989.

[A3] H. D. Chiang, F. F. Wu, P. P. Varaiya, "A BCU Method for Direct Analysis of Power System Transient Stability," IEEE/PES Summer Meeting, San Diego, July 1991, paper 91 SM 423-4 PWRS.

[A4] M. Vidyasagar, *Nonlinear Systems Analysis*, Prentice-Hall. 1978.

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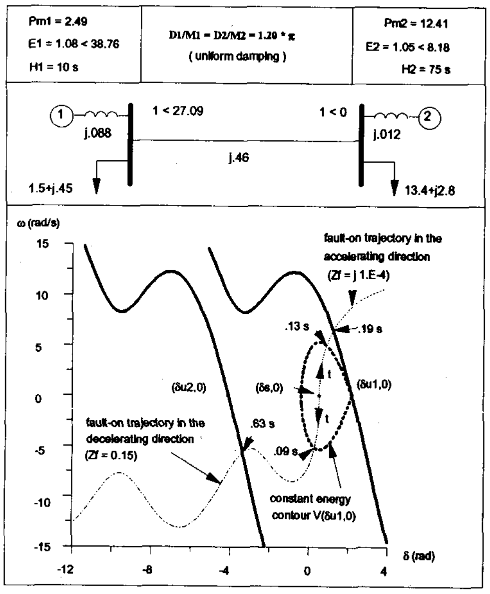
A. LLAMAS, J. DE LA REE LOPEZ, L. MILI, A. G. PHADKE, J. S. THORP: The authors would like to thank the discussers for their interest in our paper. In response to the questions raised by Messrs. Ejebe and Tong, we offer these comments: The four UEPs listed in table 1 are type-1, they lie on the stability boundary of the half order gradient system (PEBS), and they are labeled δ_1,δ_2,δ_3 and δ_4 in Figure 11. The corresponding UEPs of the original (full order) system are (δ_1,0,0),..., (δ_4,0,0). Of these four UEPs only (δ_1,0,0) belongs to the stability boundary of the original system. Clearly this power system does not satisfy the one-parameter transversality condition.

In faults (ii) and (iv) a resistive impedance (0.06 pu and 0.03 pu) is required to yield the corresponding UEPs. To expose that (δ_3,0,0) does not belong to the stability boundary of the original system, we had to use four digit accuracy in the cct (0.1955 s); a cct of 0.195 seconds yields a stable case.

Computation of the exit point of the stability boundary of the gradient system using sustained short circuit fault-on trajectory is not in the BCU articles. This is one way to assure that only UEPs in the forward direction are obtained. However, this may yield a very conservative result as illustrated in the figure.



In this case the clearing time obtained by using $(\delta_{u1},0)$ is 0.09 seconds, which is very small compared to the time when the exit point is reached: 0.63 seconds. Using $(\delta_{u2},0)$ gives a better result.

In response to Mr. Canizares, we agree that there is a critical uniform damping λ_c, such that the one-parameter transversality condition is satisfied if

$$\frac{D_i}{M_i} = \lambda \geq \lambda_c \, (i = 1, 2, \ldots.., n_{gen})$$

We believe (although have no proof for this statement) that the one-parameter transversality condition is more likely to be satisfied if uniform damping is increased as explained above, or if line loading is decreased, which tends to reduce the spread in internal machine rotor angles.

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